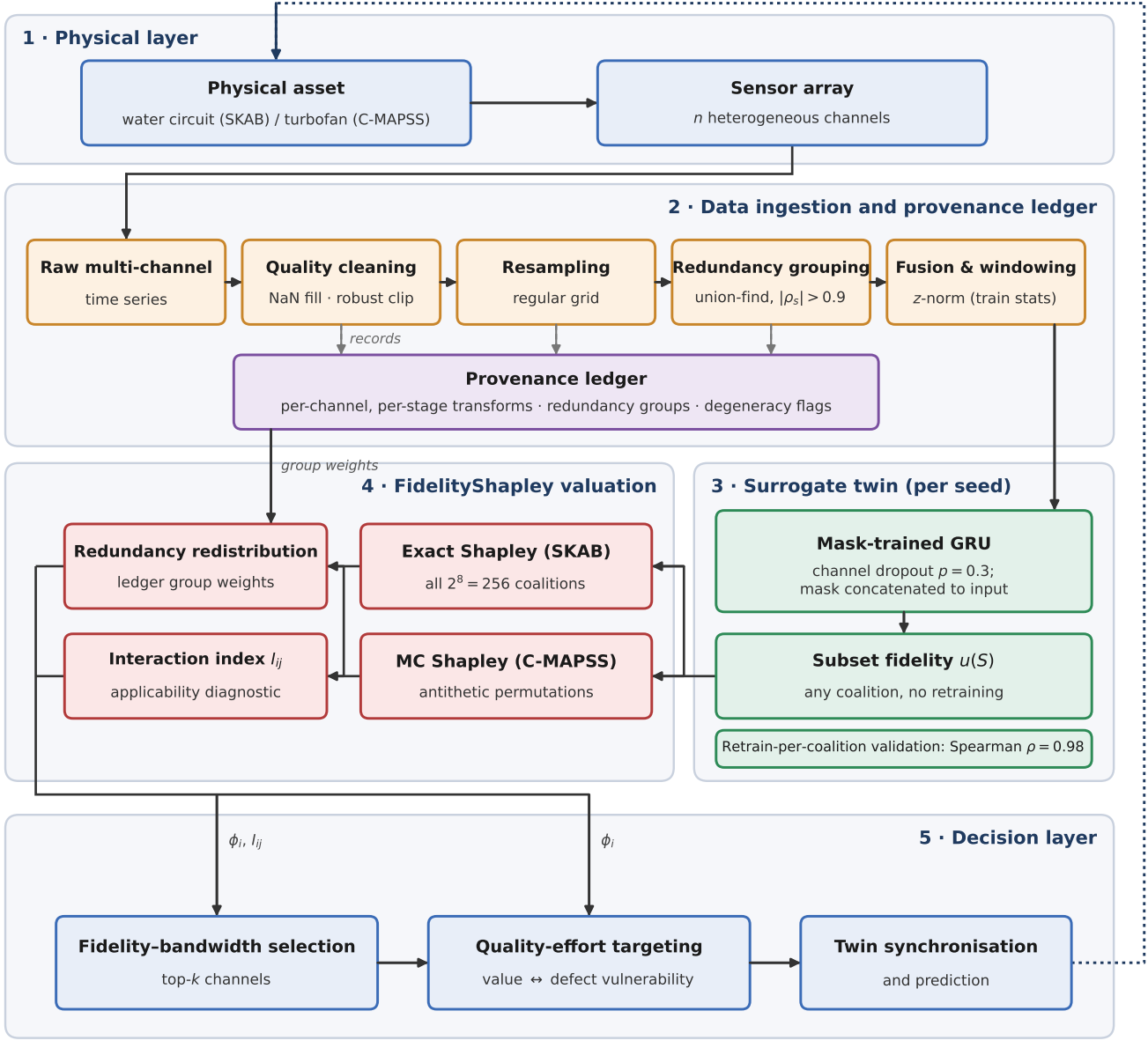
1 · Physical layer
Physical asset
water circuit (SKAB) / turbofan (C-MAPSS)
Sensor array
n heterogeneous channels
2 · Data ingestion and provenance ledger
Raw multi-channel
time series
Quality cleaning
NaN fill · robust clip
Resampling
regular grid
Redundancy grouping
union-find, |ρs| > 0.9
Fusion & windowing
z-norm (train stats)
records
Provenance ledger
per-channel, per-stage transforms · redundancy groups · degeneracy flags
group weights
4 · FidelityShapley valuation
Redundancy redistribution
ledger group weights
Exact Shapley (SKAB)
all 2^8 = 256 coalitions
Interaction index I_ij
applicability diagnostic
MC Shapley (C-MAPSS)
antithetic permutations
3 · Surrogate twin (per seed)
Mask-trained GRU
channel dropout p = 0.3;
mask concatenated to input
Subset fidelity u(S)
any coalition, no retraining
Retrain-per-coalition validation: Spearman ρ = 0.98
φ_i, I_ij
φ_i
5 · Decision layer
Fidelity–bandwidth selection
top-k channels
Quality-effort targeting
value ↔ defect vulnerability
Twin synchronisation
and prediction
closed-loop feedback